

1. A causal LTI system S with impulse response $h(t)$ has its input $x(t)$ and output $y(t)$ related through a linear constant -coefficient differential equation of the form

$$\frac{d^3 y(t)}{dt^3} + (1 + \alpha) \frac{d^2 y(t)}{dt^2} + \alpha(\alpha + 1) \frac{dy(t)}{dt} + \alpha^2 y(t) = x(t)$$

- a. If $g(t) = \frac{dh(t)}{dt} + h(t)$, then how many poles does $G(s)$ have?
 - b. For what real values of parameter α is S guaranteed to be stable?
2. Consider an LTI system with input $x(t) = e^{-t} u(t)$ and impulse response $h(t) = e^{-2t} u(t)$.
- a. Determine the Laplace transforms of $x(t)$ and $h(t)$.
 - b. Using the convolution property, determine the Laplace transform $Y(s)$ of the output $y(t)$.
 - c. From the Laplace transform of $y(t)$ as obtained in part (b), determine $y(t)$.
3. Consider an LTI system with input $x[n]$ and impulse response $h[n]$. Determine the output $y[n]$ by computing the inverse z-transform of the product of the z-transforms of the input and the unit sample response.

$$h[n] = \begin{cases} \alpha^n, & n \geq 0 \\ 0, & n < 0 \end{cases} \quad x[n] = \begin{cases} 1, & 0 \leq n \leq N-1 \\ 0, & \text{otherwise} \end{cases}$$

4. A causal LTI system is described by the difference equation given below;
- $$y[n] = y[n-1] + y[n-2] + x[n-1]$$
- a. Determine the system function $H(z)$. Plot the poles and zeros of $H(z)$ and indicate the region of convergence.
 - b. Find the unit sample response of the system.
 - c. Decide if it is stable or not.